

```

{
  "plans": [
    {
      "plan_date": "2026-05-28",
      "type": "buy",
      "stock": "æé¿¿â&#x2013;è¿¿",
      "code": "601177",
      "condition": "ä,Šæ¶"è¶¶âŠ¿Ä·ä,Šè¿¶",
      "condition_category": "ä,Šæ¶¶"è¶¶âŠ¿¿", // æ¿ä»¶â¿$ç¿»¿¿^¿"~ä°¿¿»¿è¿¿â¿¿¿»„¿¿¿
      "condition_detail": "ä,Šè¿¶", // æ¿ä»¶¿¿¿¿¿¿¿¿
      "stop_loss": 17.6,
      "stop_loss_pct": 5.22,

      "plan_close": 18.56, // è¿¿â^'â¿¿"â¿¿ç¿„æ"¶¿¿¿~ä»·
      "next_date": "2026-05-29",
      "next_open": 18.70,
      "next_close": 18.90,
      "next_high": 19.20,
      "next_low": 18.50,

      "change_pct": 1.83, // (next_close - plan_close) / plan_close * 100
      "max_gain": 3.45, // (next_high - plan_close) / plan_close * 100
      "max_loss": -0.32, // (next_low - plan_close) / plan_close * 100
      "hit_stop_loss": false, // ¿¿~ä,æ~â¿¿·¿ç 'æ¿æ¿ä»·

      "result": "success", // success | failure | flat(Ä±0.5%â¿¿...) | no_data
      "executed": null, // true | false | null(æææ¿¿¿¿¿¿¿¿¿¿"æ^·æ¿è°°)
      "user_note": "" // ¿"~æ^·â¿¿¿æ³"¿¿^ä,°ä»¿æ¹^æ^â¿¿¿/â¿¿±è'¥¿¿¿

    }
  ],
  "summary": {
    "total_plans": 7,
    "by_type": { "buy": 5, "sell": 1, "watch": 1 },
    "success": 4,
    "failure": 3,
    "flat": 0,
    "success_rate": 57.1,
    "avg_gain_pct": 2.1, // æ^â¿¿¿è¿¿â^'¿¿„â¿¿³â¿¿¿¶"â¿¿...
    "avg_loss_pct": -1.5, // â¿¿±è'¥è¿¿â^'¿¿¿„â¿¿³â¿¿¿è·¿â¿¿...
    "best_gain": 5.2,
    "worst_loss": -3.1,
    "win_loss_ratio": 1.4 // avg_gain / |avg_loss|
  },
  "by_condition": {
    "ä,Šæ¶¶"è¶¶âŠ¿¿": { "total": 4, "success": 3, "rate": 75.0 },
    "â¿¿é-éæ¿¿¿": { "total": 2, "success": 1, "rate": 50.0 },
    ...
  },
  "last_updated": "2026-05-30T10:00:00"
}

```

```
GET /api/plan-tracking
è¸"â>ž: {
  plans: PlanItem[],           // æ¸æ¸æ¸;â' + è¸½è, ºç»ªæ¸
  summary: Summary,           // ç»ªè¸;æ¸;æ¸æ¸
  by_condition: {},           // æ¸æ¸;â»ªç¸±æ¸ž<â^†ç»ª, ç»ªè¸;æ¸
  last_updated: "2026-05-30T10:00:00"
}
```

POST /api/plan-tracking/annotate

```

a, a: {
  plan_id: "2026-05-28-buy-0",
  executed: true/false,
  user_note: "ç>~ä,ë$!ä'ä¹°ç,¹i¼EaE%è@;ä'æ$ë;E"
}
è¸"ä>ž: { success: true }

```

POST /api/plan-tracking/refresh

```
èš|ǎ'éta-°è®|ç®-æ:€æ:æ:ǎ®|æ'çš,,è®|ǎ'°è:½è,ǎ
è:"ǎ>ž: { updated: 5, no_data: 1 }
```

3.5 Μέθοδος

æ°åžç<¬ç««é¡µéç `PlanTracking.tsx`ĩ¼Ééšè¿†å°•éƒ°å¬¼è^æè®¿é—®ĩ¼Èè·°å¾„, `/plan-tracking.html`ã€,

é;μέϕς»“æž,,i¼š

1. **é;íéí"ç»Yëò;â;ç;è;E**í¼"â, È;E3â, ââ...³é"®æ£±æ ±í¼%:
| æ"âSÿÇ± | â¹³â±ç»â"â | ç»â"âæ" |
| - - - - - |
| â±çâ"ç»³â"æ"æ"â"â | âæ... +2.1% | 1.4 â£ |
| æ"âSÿN/æ»N | ææ"â½³+5.2% | ææ"â"®-3.1% |

2. *æ̥%æ;ä»¶ç±»âž<â^†ç»„ç»Yè®|**ĩ¼^â-æš~â ĩ¼%:

ðŸ”ˆ^ æƒæ;æ; ä»¶ç±»äž‰
 â–¶ ä, Šæ¶”è¶¼šŽÄ·ä, Šè;Æ: 3/4=75% +2.8%
 â–¶ ä, Šæ¶”è¶¼šŽÄ·ç¾œæ†•ç†• 0/1=0% -1.2%
 â–¶ äÆœé—éœæè;â, âÆœé—äº•é†• : 1/2=50% +0.5%

3. $\mathbb{R}^n \ni x \mapsto \frac{1}{2} \|x\|^2$ is a convex function.

æ-ʏæŋ		çʊ>äʒ<		è, içʏ		æj>æŋ		æ-ʏæŋ		è.œ		ç>æʒœ		æ:ʒèjœ		äʃæʒ
05-28		buy		æéçj>äèçj>		ä.ʒœŋ		èœçj>äçj>ä.ʒèjœ		+1.83%		äœœ		æ>äçj>		äçj>æ çèœœ
05-28		buy		ä.äœœççj>æçœ		äœœé-éœçj>ä.äœœäœ		-2.10%		äœ		äçj>æ çèœœ		äçj>æ çèœœ		
05-26		buy		äçj>æ çèœœ		ä.ççj>äçj>ç, 1		+0.55%		äçj>		äçj>æ çèœœ		äçj>æ çèœœ		

4. $\ast \mathring{a}^{\circ} \bullet \acute{e} \acute{f} \ddot{x} \ddot{a}^{\frac{1}{2}} \ddot{x} \ast \mathring{i}^{\frac{1}{4}} \mathring{s} \mathring{a}^{\wedge} \cdot x^{\circ} x^{\circ} x^{\circ} \mathring{a} \mathring{e} \mathring{a}^{\circ} \bullet \acute{e} \acute{f} \ddot{a}^{-\frac{1}{4}} \acute{e}^{\wedge} \mathring{a}$

5. **æ- è®;ă^ 'æ-ŋçš,,çšŋæç**i¼šçº-æ-†ă-æçºº"æš,æ- è®;ă^ 'æ-ºæ®i¼çè-·ă...^ăæ" [ă·¥ă¼æºº] (/workbench.html)ă^ŋă®šæ~Žă-¥è®;ă^ "

3.6 æ—¥æœŸèƒå›‘ç›é€%

ç" "æ^ · â - è ‡ ¢ ç" ± € % æ < © ç » ÿ è ® ; ç š „ æ - ¶ é - ' è ¶ f â > ' ï ¼ ¶ å ž ç « - å ... é ‡ å ~ å , " ï ¼ ¶ å ž ç « - æ ¶ æ - ¥ æ æ ÿ è ¿ ‡ æ » ¢ ã € ,

```
``json
// GET /api/plan-tracking?start_date=2026-05-01&end_date=2026-05-30
``
```

****è§„ä¸ªTMï¼****

[illegible]

****ç•Œéçï¼š****

[illegible]

3.7 è†ªåŠ¨å»ºè®®ç³»»ç»™ÿ¼^v1ï¼¸

åŽç«-åœ¨è¿"å>žè¿½è, æ•°æ®æ-¶i¼£è‡æš-å^†æžç»“æžœå¹¶ç”Yæ^å-æ“ä½œæ»ºè®®æ€,

****â»ºè®®ès,,â ^™ï¼š****

èš„â[™]	èš;â'æ;ä»¶	âºè©çª²ä¼<
æ;ä»¶æ^âšŸçžž+ääžž	æŸæ;ä»¶âššç±æ^âšŸçžž < 50% ä,"æ¬;æ° => 3	âš ì, â...³æ³..î¼šâ€œç¼œé†æ'ç†â€œ3æ¬;ä»¶...
33æ^âšŸçžž+î¼œä½žäºžæ^'ä½"		
æ;ä»¶æ^âšŸçžž+æ¶â€-	æŸæ;ä»¶è¿'Næ¬;æ^âšŸçžžæ^-â...ëf^âž†âž†ä,'<é™ > 15%	âš ì, æ¶â€-î¼šâ€œâ€œâ€œé-'éæ†è;â€œì'10æ¬;40% vs â...
^ëf^57%î¼œä,'<é™17%		
æ;ä»¶æ¶è^-çžž+ä½žçš€	æŸæ;ä»¶æ^âšŸçžž > 70% ä,"æ¬;æ° => 5	ðŸ'; ææä½ž²î¼šâ€œä,Sæ¶^è¶&âšžâ-ä,Sè;äâ€œ8æ¬;88æ^âšŸçžž+î¼œç»šç»-
äžæ |

| æƧæŸëœ¼ç½œðç´š | ç>¨ä,è·Œç ´æƧæŸçš,,æ¨"ä¼< > 25% | ðŸ¨Š æƧæŸäç´ši¼š12ç¨"ä,4ç¨"ç>¨ä,æƧæŸëç«èš¡ä´i¼^33%i¼%i¼Œä»ºèœœ"¼äœ½ | ä,äè,¡éç´ç¹ä±è´Ÿ | äŒä,€è,¡çŸ¹ä±è´Ÿæ¡¡œ¨ >= 3 | äš i. ä,äè,¡i¼šXXè,¡çŸ¨5æ¡¡èœ¡ä´ä,4æ¡¡ä±è´Ÿi¼Œæ³¨æ,,è¨†ä^« |

```
**ä»ºèœœ¨¨æ¨æœç»“æž,,i¼š**
```json
{
 "suggestions": [
 {
 "type": "warning", // warning | improving | best | info
 "dimension": "condition", // condition | stock | stop_loss | type
 "category": "ä,šæŸ¨¨èŸ<äš¿ä·ç¼œé†æ¨´ç†",
 "rate_current": 33.3,
 "rate_overall": 62.5,
 "count": 3,
 "message": "æ¡ä»ŸŸäŒŒç¼œé†æ¨´ç†ä€3æ¡¡ä»...33%æ¨äšŸçžž†i¼^æ¨´ä½“62.5%i¼%i¼Œä»ºèœœæfçæŸŸè¨Ÿæ¡¡ä»ŸŸç>éç%éç»è¼´ ",
 "action": "check_condition" // éç,,ç¨i¼ššä%ç«¨ä¨ç,¹ä†»è¨³è½¨
 }
]
}
```

```
æ%šè¡Œæ¨Ÿæœºi¼š æ¨æ¡¡ `compute_tracking()` æ¨æ¨ `get_tracking()` æ¨Ÿè†ºäš¨ç¨Ÿæ¨äç,

```

## 4. ä,äšši¼^v1èŒfä>´ä±-i¼%

é¡¹ç>œ	äžŸä>
ç¨¨3ä±œ/ç¨¨5ä±œèè½è,ä	ç¨¨æ¨·æ¨žç¡œä...^çœ<ç¨¨2ä±œ
è†äšš¨æ †èœ¨"ä,²æ%šè¡Œ"	éœçèèæžŸä...Ÿäºæ¨¨ç³»ç»Ÿi¼ŒœŸ...äººv1èŒfä>´
äœžç>æ¨äºæä»·è½è,ä	æ²¡æœ%ä^,ä¨†APIæžŸä...Ÿ
watch/èš,ä¨Ÿèœ¡ä¨´çŸŸèœ¡	èš,ä¨Ÿèœ¡ä¨´ä,ç¡œäºšæ¨ä½œæ¨¹ä´
æ¿ä-/ä±šç>¨äŒæŸä¨¹æ¨"	v2ä†æ%ä±¨
ä>¼è¡¨ä¨èš†äŒ-i¼^Kçº¿æ †æ³¨i¼%	æ¨†ææ¨è¡¨æ ¼èŸºä±Ÿi¼Œv2ä¨äš
è†äšš¨ä¼¨äŒ-ç³»ç»Ÿä,æ¨º	v2è†äšš¨ä»ºèœœä...^äšši¼Œäººäššä†³ç¨

## 5. v2èš,,ä¨´i¼ššè†äšš¨è¿>äŒ-ç³»ç»ŸŸ

### 5.1 ç>œæ †

ä»ž¨äººçœæ¨æ¨æºäšä¼¨äŒ-¨ä†çºšä,º"æœºä¨¨ä¨†æžä†´ç¨¨ä†äº»ºèœœä†äººç¡œèœœä†´ç³»ç»Ÿè†äšš¨èºfæ¨´"çš,,é-çž¨äç,

### 5.2 æ¼¨¨è¿>è¨ä¼,,

```
```
v1i¼^ä½"ä%í¼%: æ¨ºæœä±¨ç±º + æœºä¨¨ä»ºèœœi¼^äººä·Ÿä†³ç-i¼%
      ä†"
v2:      æœºä¨¨ä»ºèœœ ä†´ äººä·Ÿç¡œèœæ ä†´ ä,æ¨ºè†äšš¨èºfæ¨´
      ä†"
v3:      æœºä¨¨ä»ºèœœ ä†´ è†äššš¨èºfæ¨´i¼^äººä·Ÿä¨ä>žæ»šší¼%
```
```

### 5.3 v2æ<Ÿäœžçž¨äššŸèf½

äšŸèf½	è¨æ¨ž
\*\*æ¡ä»ŸŸä,æ¨ºä»ºèœœ\*\*	ä¡,æžœæŸæ¡ä»ŸŸæ¨äššŸçžž†æŒç<40%i¼Œä»ºèœœä¿œ¨¹èŸæ¡ä»Ÿçš,,ä¹ºç,¹ä,æ¨ºæ¨-äœç¨¨
\*\*æƧæŸë¨^äç¼ää»ºèœœ\*\*	äŸºäºžäž†ä²æƧæŸèš¡ä´çžž†i¼Œä»ºèœœæ¨¼äœ½æ¨-æ¨Ÿç´šæƧæŸä¹ºº¡
\*\*ä¹ºç,¹ä¿¡ä·ä»ºèœœ\*\*	ä¨äº>ä¹ºç,¹ä¿¡ä·ç»¼ä¨è¡¨çžºäŸ½/ä·œi¼Œä»ºèœœèºfæ¨´ä¿¡ä·æfè†
\*\*ä,äè,¡é¨ää¨\*\*	é¿æœŸä±è´Ÿçžž†«¨çš,,è,¡çŸ¨è†äšš¨æ †çºŒæè†
\*\*è†äššš¨ä>žæµ<¨\*\*	ä¨¹ä»ºèœœçš,,æ¨¹äšš¨è¿>è¡Œäž†ä²ä>žæµ<i¼Œç»¨ä†ºä¡,æžœæ¨¹æ¨Xi¼Œäžž†ä²æ¨äššŸçžžä¼šä¨æ¨Ÿ%
\*\*æ¨¹è¿>æšŸä¨š\*\*	æ¨ä¨/æœ¨èè†äšš¨ç¨Ÿæ¨ç³»ç»Ÿä¼¨äŒ-ä»ºèœœæšŸä´ší¼^PDFi¼%

### 5.4 æ¨ºæœä¼èµ-

v2ä»ºèœœä,ä†ä»...ä¼èµ-èœ¡ä¨´è¿è,äæ¨ºæi¼ŒŒè¿éœçè¡i¼š  
- ä,äè,¡äœŒæ¨´Kçº¿i¼^ä>žæµ<ç¨¨i¼%  
- ä¹ºç,¹ä¿¡ä·ä¨æ-ç»æžœ  
- æƧæŸæ¨äºææ¨ºæi¼^ä¡,æžœæžŸä...Ÿäœžç>¨i¼%

### 5.5 æšçææ¨æ¹æ¡^

```
```python
# ä»ºèœœä¼·æ¨ži¼^ä¼ää»fç i¼%
class SuggestionEngine:
    def analyze(self, plans, klines):
        suggestions = []
        # 1. æ¡ä»ŸŸè¡¨çžž¨ä†æž
        for condition, stats in self._group_by_condition(plans):
            if stats.success_rate < 0.4:
                # æ¨¡æ<Ÿèºfæ¨´ä,æ¨ºäžçš,,è¡¨çžº
                backtest_result = self._backtest_new_params(condition, klines)
```

- âˆ†æ™ˆˆˆ: ˆ feature/plan-tracking`
- âŸŸäºŸ master âˆ†ˆâ»º
- ä...^èˆ™ä ˆè®äè®:æ-æ:æf + PDF ä®èˆ...
- ç„Ÿäšš TDD ä¼€ä'
- ä®æˆˆâššäˆ¹äšš master